

Kuan-Wen Lai

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Empolymnt & positions

National Center for Theoretical Science, Taiwan
Member affiliated with the Mathematics Division, 2025.03–present
Tunghai University, Taiwan
Assistant Professor, 2024.08.01–present
Academia Sinica, Taiwan
Postdoctoral Researcher, 2023.09.01–2024.07.31
Mentor: Yuan-Pin Lee
Universität Bonn, Deutschland
Postdoctoral Researcher, 1 September 2021–31 August 2023
Mentor: Daniel Huybrechts / Part of the ERC Synergy Grant [HyperK \(10.3030/854361\)](#).
University of Massachusetts Amherst, USA
Visiting Assistant Professor, September 1, 2018–August 31, 2021
Mentor: Eyal Markman
R.O.C. (Taiwan) Army, Taiwan
Obligatory military service, 2011.08.08–2012.07.09

Education

Brown University, USA
Ph.D. Mathematics, May 27, 2018
Thesis: *Cremona transformations and rational parametrizations inspired by Hodge theory*.
Advisor: Brendan Hassett
Rice University, USA
Graduate student, August 2013–July 2015
Transferred to Brown University with thesis advisor Brendan Hassett.
Taida Institute for Mathematical Sciences, Taiwan
Research Trainee, 2012.08–2013.07
Mentor: Chin-Lung Wang
National Taiwan University, Taiwan
M.S. Mathematics, 2011.06.04
Thesis: *The Gromov-Witten Theory of $\mathbb{P}^1$* .
Advisor: Hui-Wen Lin
B.S. Physics, with minor in Mathematics, 2009.06.07

Grants

- Academic Summit Program 114-2639-M-002-009-ASP (CI), with Jung-Kai Chen (PI), Hsueh-Yung Lin, Jheng-Jie Chen, and Ching-Jui Lai, 2025.08.01–2030.07.31
- NSTC Research Project 113-2115-M-029-003-MY3 (PI), 2024.10.01–2027.07.31

Research

My research lies in algebraic geometry, with a particular focus on Fano fourfolds with K3-type Hodge structures, as well as on Cremona transformations from the perspective of derived categories of coherent sheaves. I have also worked on algebraic geometry over finite fields and its applications to coding theory.

Preprints

- A geometric approach to the density of rank-metric codes (with Shamil Asgarli, Lian Duan, and Nathan Kaplan), in preparation.
- Mirror symmetry for lattice-polarized abelian surfaces (with Yu-Wei Fan), <https://arxiv.org/abs/2602.04232>.
- Categorical Nielsen realization problems for generic K3 surfaces (with Yu-Wei Fan), <https://arxiv.org/abs/2302.12663>.

Publications / Accepted papers

1. Fourier–Mukai numbers of K3 categories of very general special cubic fourfolds (with Yu-Wei Fan), [Israel Journal of Mathematics](#) 268, 427–437 (2025).
2. On the irrationality of moduli spaces of projective hyperkähler manifolds (with Daniele Agostini and Ignacio Barros), [Épjournal de Géométrie Algébrique](#), Volume 9 (2025), Article No. 13.
3. Frobenius nonclassical hypersurfaces (with Shamil Asgarli and Lian Duan), [Mathematische Zeitschrift](#) 309, 18 (2025).
4. Transverse linear subspaces to hypersurfaces over finite fields (with Shamil Asgarli and Lian Duan), [Finite Fields and Their Applications](#) 95 (2024), Paper No. 102396.
5. New rational cubic fourfolds arising from Cremona transformations (with Yu-Wei Fan), [Algebraic Geometry](#) 10 (4) (2023), 432–460.
6. On the irrationality of moduli spaces of K3 surfaces (with Daniele Agostini and Ignacio Barros), [Transactions of the American Mathematical Society](#) 376 (2023), No. 2, 1407–1426.
7. Uniform potential density for rational points on algebraic groups and elliptic K3 surfaces (with Masahiro Nakahara), [International Mathematics Research Notices](#) 2022, no. 23, 18541–18588.
8. Decompositions of Lagrangian classes on K3 surfaces (with Yu-Shen Lin and Luca Schaffler), [Mathematical Research Letters](#) 28 (2021), no. 6, 1739–1763.
9. Bijective Cremona transformations of the plane (with Shamil Asgarli, Masahiro Nakahara, and Susanna Zimmermann), [Selecta Mathematica, New Series](#) 28, 53 (2022).
10. Transverse lines to surfaces over finite fields (with Shamil Asgarli and Lian Duan), [manuscripta mathematica](#) 165, 135–157 (2021).

11. Cremona transformations and derived equivalences of K3 surfaces (with Brendan Hassett), [Compositio Mathematica 154 \(2018\), Issue 7, 1508–1533](#).
12. New cubic fourfolds with odd degree unirational parametrizations, [Algebra & Number Theory 11 \(2017\), No. 7, 1597–1626](#).

Proceedings

- A tour of the rationality problem of cubic fourfolds, Proceedings of the International Congress of Chinese Mathematicians 2020.

Invited talks

- 2027 Taiwanese Mathematical Society (TMS) Annual Meeting, National Taiwan University, January 18–19
- 2026 Mini-Workshop: Algebraic Geometry in Xitou, July 19–21
 - 2026 Summer School in Algebraic Geometry, National Center for Theoretical Sciences (NCTS), June 29–July 17
 - Tour de Formosa: Algebraic Geometry along Taiwan’s eastern coast, January 12–16
- 2024 Algebraic Geometry in Taipei, National Taiwan University, December 2–6
 - Colloquium, National Sun Yat-Sen University, March 8
 - Colloquium, National Kaohsiung Normal University, March 7
 - Taiwanese Mathematical Society (TMS) Annual Meeting, National Chengchi University, January 22–23
 - Taipei Postdoc Seminar, Academia Sinica & National Center for Theoretical Sciences (NCTS), January 3
- 2023 Student Geometry Seminar, National Tsing Hua University, November 16
 - Algebraic Geometry in East Asia, Korea Institute for Advanced Study (KIAS), November 6–10
 - NCTS Seminar in Algebraic Geometry, National Center for Theoretical Sciences (NCTS), September 22
 - Algebra, geometry and number theory seminar, Universiteit Leiden, April 26
 - Higher Dimensional Algebraic Geometry Workshop, National Center for Theoretical Sciences (NCTS), March 20–24
 - Colloquium, National Central University, March 16
 - Colloquium, National Cheng Kung University, March 9
- 2022 Birational Complexity of Algebraic Varieties, Simons Center for Geometry and Physics (SCGP), December 12–16
 - Géométrie Algébrique et Géométrie Complexe, Centre International de Rencontres Mathématiques (CIRM), November 28–December 2
 - Algebraic and Arithmetic Geometry Seminar, Universität Bielefeld, June 1
 - Colloquium, National Tsing Hua University, March 21 (Online)
- 2021 Arithmétique et géométrie algébrique, Université de Strasbourg, December 9

- A tour of the rationality problem of cubic fourfolds, Academia Sinica, Taipei, Lecture series during July 28–August 3
- Seminar of Algebraic Geometry in East Asia, April 16 (Online)
- 2020 International Congress of Chinese Mathematicians (ICCM) at the University of Science and Technology of China, December 27–29 (Online)
- Mathematical Physics Seminar, Center of Mathematical Sciences and Applications (CM-SA), April 20 (Online)
- Geometry, Physics, and Representation Theory Seminar, Northeastern University, January 23
- Colloquium, National Central University, Taiwan, January 9
- 2019 Front Range Algebraic Geometry and Number Theory (FRAGMENT) seminar, Colorado State University, December 12
- Algebra and Number Theory Seminar, Dartmouth College, September 17
- A Summer Day on Algebraic Geometry, Center for Advanced Study in Theoretical Sciences (CASTS), July 1
- NT & AG Seminar, Boston College, February 7
- Workshop on Moduli Spaces and Related Problems, Shanghai Center for Mathematical Sciences (SCMS), September 17–21
- Valley Geometry Seminar, UMass Amherst, March 2
- 2017 Mini-talk at Georgia Algebraic Geometry Symposium (GAGS), University of Georgia, March 3–5
- Yale Algebraic and Tropical Geometry, Yale University, February 16
- Princeton Algebraic Geometry Seminar, Princeton University, February 14
- Algebra and Algebraic Geometry Seminars, Brown University, February 10
- 2016 Algebra and Algebraic Geometry Seminars, Brown University, October 14
- 2013 Algebraic Geometry and Number Theory Seminar, Rice University, December 3

Teaching & related activities

Tunghai University

2026 115-1 Asobou! Combinatorial Games

- 115-1 Calculus A (I)
- 115-1 Mission Mars (II)
- 114-2 Differential Geometry
- 114-2 Calculus A (II)
- 114-2 Undergraduate Research (9 students)
- 114-2 Mission Mars (I)
 - A course in collaboration with a science fair, as part of the "Mission Mars" event at the National Museum of Natural Science in November each year.

2025 114-1 Asobou! Combinatorial Games

- 114-1 Calculus A (I)
- 114-1 Mission Mars (II)
- 113-2 Calculus A (II) (Two sections)
- 113-2 Undergraduate Research (6 students)
- 113-2 Mission Mars (I)
- UTAR $\times$ Tunghai International-National Mathematics Competition
 - A math competition in July 2025 co-organized by my department and the UTAR Mathematics Society of Universiti Tunku Abdul Rahman (UTAR).

2024 113-1 Calculus A (I) (Two sections)

- 113-1 Mission Mars (II)

Universität Bonn

2022 Summer Semester, Birational Geometry of Algebraic Varieties

UMass Amherst

2021 Spring, Fundamental Concepts of Mathematics (Two sections)

2020 Fall, Independent Study on Algebraic Geometry

- Advisee: Andrew Tawfeek (Amherst College)
- Fall, Fundamental Concepts of Mathematics (Two sections)
- Spring, Independent Study on Algebraic Geometry
 - Advisee: Andrew Tawfeek (Amherst College)
- Spring, Fundamental Concepts of Mathematics
- Spring, Multivariate Calculus

2019 Fall, Multivariate Calculus (Two sections)

- Spring, Calculus II (Two sections)

2018 Fall, Calculus I (Two sections)

Brown University

2017 Fall, Intermediate Calculus

2015 Fall, Introductory Calculus I

Rice University

2015 Summer, Single Variable Calculus II